FORM TP 2012023

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

PHYSICS

Paper 032 – General Proficiency

Alternative to SBA

2 hours 10 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. You MUST use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.
2. ALL WORKING MUST BE SHOWN in this booklet, since marks will be awarded for correct steps in calculations.
3. Attempt ALL questions.
4. The use of silent, non-programmable calculators is permitted.
5. Mathematical tables are provided.
6. You are advised to take some time to read through the paper and plan your answers.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Attempt ALL questions.

You MUST write your answers in this answer booklet.

1. Sandy was performing an experiment to determine the refractive index of a glass block using Snell's Law. Her results are shown in Table 1.

TABLE 1: RESULTS OF EXPERIMENT

Angle of Incidence, *i*/°	**Angle of Refraction, *r*/°**	**sin î**	**sin r̂**
10.0	6.0	0.174	0.105
	16.0		0.276
30.0	20.0	0.500	0.342
40.0	24.0	0.643	0.407
50.0	30.0	0.766	0.500

(a) A protractor is shown in Figure 1. Complete Table 1 by reading the protractor to measure the angle of incidence, i, and enter sin î. **(2 marks)**

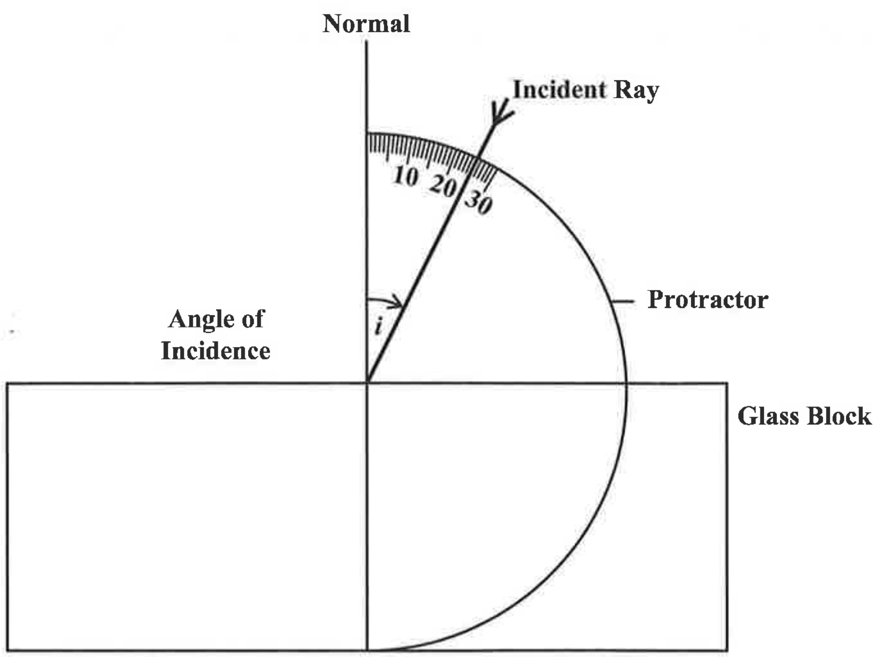

Figure 1. Protractor measurement for angle

(b) State TWO precautions Sandy should take when conducting the experiment.

(2 marks)

(c) Using a suitable scale on each axis, plot a graph of sin r̂ against sin î on the graph paper on page 5. Start each axis at 0.100. **(8 marks)**

(d) Determine the gradient of the graph.

(4 marks)

(e) Use the gradient to calculate the refractive index of the glass block.

(4 marks)

Total 20 marks

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2. In an experiment to determine the I – V characteristics of an electrical component **X**, the following apparatus were used:

- A d.c. supply
- An ammeter
- A switch
- A variable resistor
- A voltmeter
- Component **X**

(a) Complete the circuit shown in Figure 2 using the apparatus listed above.

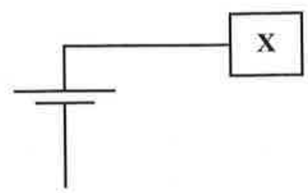

Figure 2. Circuit to determine I – V characteristics

(4 marks)

(b) By adjusting the variable resistor, the pairs of values shown in Table 2 were obtained.

TABLE 2: I – V results

Current I/A	**Voltage V/V**	**Resistance R/Ω**
0.00	0.00	————
0.27	0.10	
0.47	0.20	
0.63	0.30	
0.78	0.40	
0.90	0.50	
1.10	0.60	
1.16	0.70	
1.19	0.80	

Complete Table 2 above by calculating the eight resistance values. **(6 marks)**

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(c) On the graph paper provided below, plot a graph of I/A against V/V. **(8 marks)**

(d) Based on your results, suggest a suitable name for the component **X**.

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(1 mark)

Total 19 marks

3. A teacher projected a bead horizontally with a speed of u cms^{-1}. The bead travels along the wire shown in Figure 3 on page 9. By observing the horizontal and vertical distances from the point of projection, he proposed to determine the acceleration of the bead.

(a) Record the horizontal (x) and vertical (y) distances of the bead at the points indicated in Table 3 below.

TABLE 3: RESULTS OF THE DISTANCE OF THE BEAD

Point	**A**	**B**	**C**	**D**	**E**
x / cm					
y / cm					

(5 marks)

(b) Given that $y = \frac{a}{2u^2} x^2$, outline the steps the teacher would use to determine the acceleration, a, of the bead using a graph.

(4 marks)

Total 9 marks

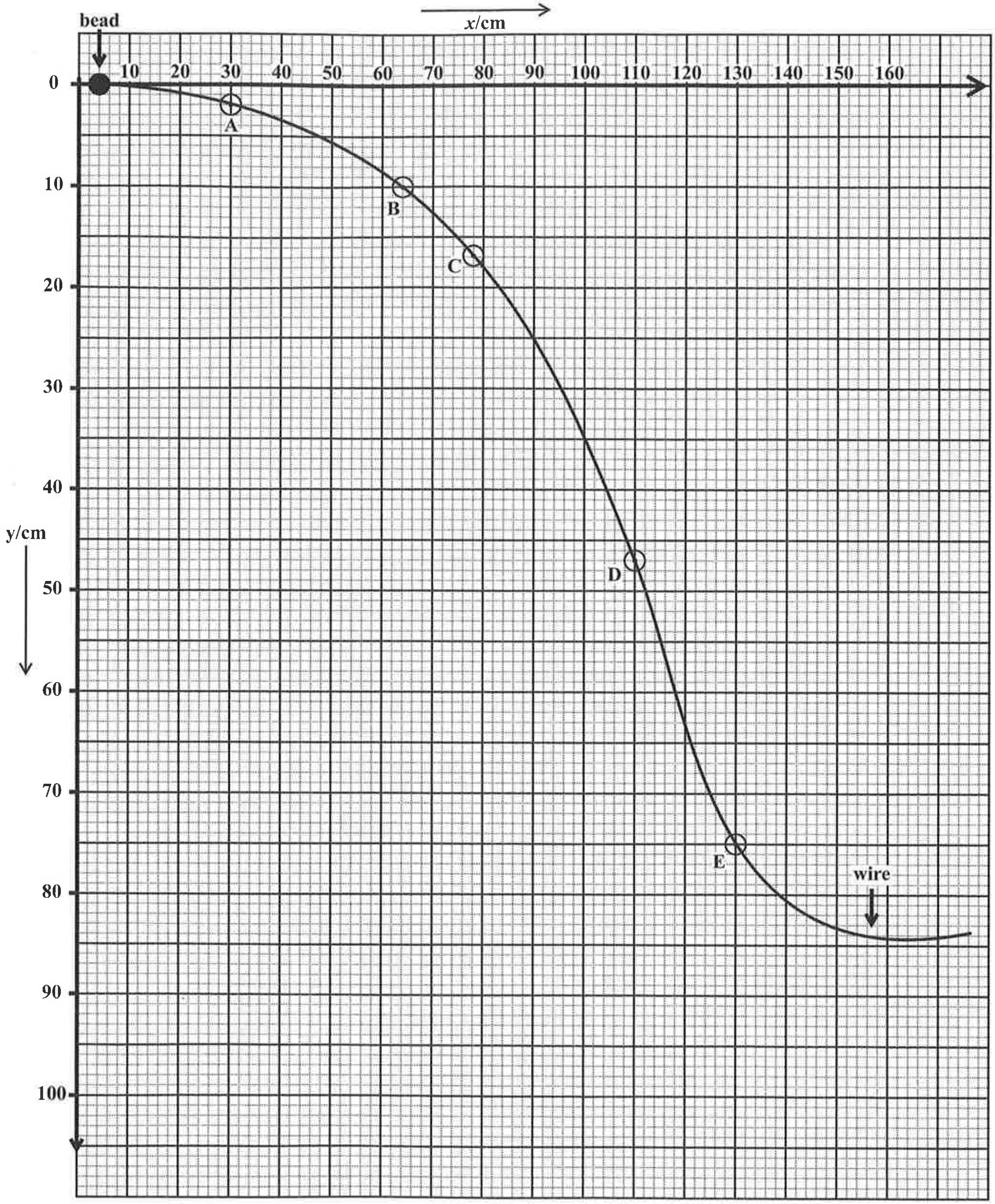

Figure 3. Position of bead after projection

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.